arohak

Purpose. People. Innovation

QUANTUM-INSPIRED OPTIMISATION · 112 USE CASE

Faster emergency response, solved on an Ising machine

Ambulance dispatch modelled as a QUBO and solved with light

In partnership with **Quanfluence**

PILOT RESULTS · SIMULATION ON REAL 108 CALL DATA

13 min sooner

to reach the patient

Current 108 (live)	22.6 min
Ising machine (sim.)	9.8 min · −57%

13 min sooner

into hospital care

Current 108 (live)	60.6 min
Ising machine (sim.)	47.5 min · −22%

How it works

from distress call to dispatch plan

Calls & ambulances
Calls, locations, routes

→

Arohak QUBO model
Rules become penalties

→

Split by zone
Each piece sized to fit the hardware

→

Quanfluence Ising machine
Each spin is a light pulse in a fibre loop

→

Dispatch plan
Right ambulance, fastest route

BUILT TO EXTEND

The model can include rules such as matching the patient to the right hospital.

ROADMAP

✓ **Simulation on 108 call data**

108 **Live ambulance trial**

112 **Police · Fire · Ambulance**

Quantum-inspired today. Quantum-ready tomorrow.

Current 108: live data, average minutes per call. Ising machine: simulation on the same 108 calls, with 15 min at the scene and 15 min hospital handover added to the simulated side only. Not yet deployed live.